



A Systematic Review for Stock Price Prediction

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Abstract—Stock price prediction has become an increasingly important research domain due to the high volatility and non-linear characteristics of financial markets. The rapid development of artificial intelligence (AI), particularly machine learning (ML) and deep learning (DL), has provided new opportunities to improve forecasting accuracy compared to traditional statistical approaches. This study presents a systematic literature review (SLR) of stock price prediction research published between 2020 and 2025, following the PRISMA 2020 guidelines. Articles were retrieved exclusively from the Scopus database and filtered using predefined inclusion and exclusion criteria, resulting in 13 relevant studies for in-depth analysis. The review examines three primary aspects: (1) the most commonly applied AI methods for stock price prediction, (2) the data sources used as model inputs, and (3) the performance evaluation metrics employed. The findings indicate that time-series-based models, particularly Long Short-Term Memory (LSTM) networks and hybrid approaches combining statistical and deep learning techniques, dominate recent research. Yahoo Finance is the most frequently used data source, while performance evaluation relies primarily on error-based metrics such as RMSE and MAE. Overall, this review highlights current research trends, methodological preferences, and existing gaps, providing valuable insights for future studies in AI-based stock market prediction.

Keywords: SLR; Stock Price; Prediction;

1. INTRODUCTION

Stock price prediction using AI methods has emerged as a significant area of research due to the inherent complexity and volatility of financial markets. The integration of artificial intelligence (AI) and machine learning (ML) techniques offers a promising approach to enhancing the accuracy of stock market forecasts. AI's ability to process large datasets rapidly and discern complex patterns makes it superior to traditional analytical methods, which often struggle with the nonlinearity and dynamics of stock prices. (Al-Alawi & Alshakhoori, 2024; Pillai & Bi, 2024).

Various AI techniques, including neural networks, support vector machines, and deep learning models such as Long Short-Term Memory (LSTM), have been employed to predict stock prices by analyzing historical data, technical indicators, and sentiment from news and social media. (Gangthade, 2024). Hybrid models that combine multiple AI techniques have been shown to outperform traditional methods by effectively handling non-linear complexities and temporal dynamics (Al-Alawi & Alshakhoori, 2024; Sun, 2025).

Despite these advancements, challenges remain, such as integrating qualitative factors like a company's strategic decisions and the impact of unforeseen events like pandemics or geopolitical tensions, which are difficult to quantify and incorporate into predictive models. (Whig et al., 2024). Moreover, the quality of data and model interpretability remain ongoing concerns that researchers aim to address by developing more sophisticated models and incorporating cutting-edge technologies such as reinforcement learning. (Sun, 2025).

The continuous evolution of AI in stock market prediction not only aids investors in making informed decisions but also contributes to the broader financial industry by enhancing investment strategies and risk management practices. (Al-Alawi & Alshakhoori, 2024; Arora et al., 2024). As AI technologies advance, their role in stock market prediction is expected to grow, offering more refined and reliable forecasting tools for investors and financial analysts alike. (Gong, 2022).

2. RESEARCH METHODS

The method used in this literature review study is the systematic literature review. This review was conducted in accordance with the PRISMA 2020 Statement. (Page, McKenzie, et al., 2021), which comprises a checklist and a flow diagram. The PRISMA Elaboration and Explanation (Page, Moher, et al., 2021) Enhances the understanding, usage, and dissemination of the PRISMA 2020 Statement by providing examples and explanations for each checklist item.

To carry out the meta-review, we adhered to the PRISMA 2020 guidelines. (Page, McKenzie, et al., 2021) Moreover, searched for pertinent systematic reviews on the prediction of stock markets using AI technologies, including ML and DL methods. The first step in this methodical strategy was to look for helpful scholarly articles in three well-known databases: SCOPUS. We employed inclusion and exclusion criteria to screen the retrieved records and selected ten papers for the final dataset. Subsequently, we extracted and coded the data from the included reviews and analyzed them from four essential perspectives: predictor methodologies, information sources, predictive model performance metrics, limitations, and future recommendations. In addition to following a well-defined research methodology, specific criteria must be met for selecting primary publications for this work.



2.1 Literature Selection Criteria

Our research phase involved retrieving relevant publications for the systematic review and selecting papers based on predetermined criteria. The strategic selection of databases further emphasized the importance of this phase, careful selection of search terms, effective use of filtering tools, and a rigorous review process. Databases were strategically selected to capture the broad scope of literature relevant to our research topic. Therefore, the Scopus database was used to retrieve papers, given its reputation as a respected academic reference and its coverage of a diverse range of high-quality scholarly articles (Wang et al., 2016).

Keywords used included: "*Stock Price*," "*Prediction*," "*Stock Market*," and "*Stock*". The articles reviewed were limited to the period 2020-2025 to ensure they remain current and relevant. Therefore, the initial stage should include several questions to be answered after the systematic review, as well as the inclusion, exclusion, and quality criteria. The research questions expected to be answered by this systematic review are presented in Table 1. The inclusion and exclusion criteria are presented in Table 2.

Table 1. Research Questions

ID	Research Questions (RQ)
RQ1	Which AI methods and technologies are most commonly used to predict stock market prices?
RQ2	Which information sources are most often used to predict stock market prices?
RQ3	Which metrics are most commonly used to verify the performance of a predictive model?

Table 2. Inclusion and Exclusion Criteria

Criterion	Inclusion	Exclusion
Year of publication	2020-2025	In addition to 2019-2025
Language	English	Non-English
Subject area	Computer Science	Non-Computer Science
Document type	Article and Conference	Non-Articles and Conference
Country/region	Indonesia	Non-Indonesia
Source type	Journal	Non-Journal
Open access	Yes	No

2.2 Research Stages

At this stage, the researcher began searching for articles related to stock prices. Related articles will be filtered according to Table 2. Based on the results of the article search, the initial identification stage yielded 3279 articles. Furthermore, after the identification stage was completed, 25 articles were selected, of which seven did not match the AI topic. From this process, 18 articles were produced, which were then filtered again because they did not use the AI method; 5 were excluded. The final result of this stage was 13 articles that met the requirements for this systematic literature review. The stages of the article search used in the systematic literature review are shown in Figure 1.

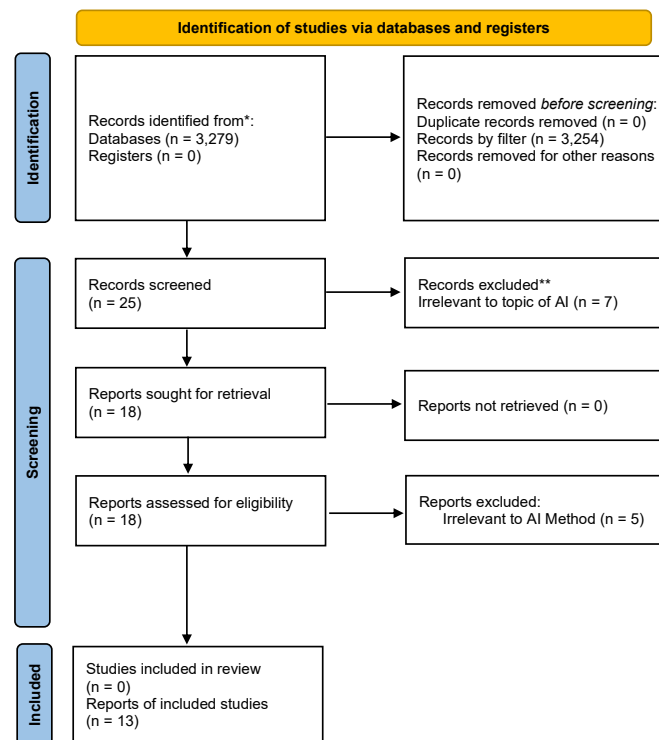


Figure 1. The number of articles separated during PRISMA

3. RESULTS AND DISCUSSION

The articles selected and reviewed by the author were then categorized according to predefined keywords. To facilitate this process, the author created a characteristics table based on the research findings. The characteristics table is shown in Table 3.

Table 3. Review the characteristics of the analyzed articles

Reference	Aim of Review	Methods	Date range of included studies
(Hansun & Young, 2021)	This study aims to apply Long Short-Term Memory (LSTM) networks for stock price prediction, focusing on stocks included in the LQ45 index. The review also discusses various forecasting methods for modeling stock price movements.	LSTM, PCA	2021
(Tarno et al., 2021)	This review assesses risk measurement using Value-at-Risk (VaR), focusing on modeling stock return volatility with ARCH/GARCH models. It also evaluates the ARIMA-GARCH approach for effective risk prediction.	ARIMA-GARCH	2012-2019
(Hoyyi et al., 2021)	This review evaluates the Variance Gamma (VG) model in finance by addressing the limitations	GBM	2019-2020



	of the Black–Scholes–Merton model and applying VG to stock price prediction.		
(Budiharto, 2021)	This review analyzes big data using R and LSTM for stock forecasting, with a particular focus on short-term data selection to improve prediction accuracy. It also highlights the importance of time series analysis in financial forecasting.	LSTM, ARIMA	No mention.
(Ahmar et al., 2021)	This review aims to identify suitable models for predicting stock prices in BRIC countries by comparing the effectiveness of the ARIMA, SutteARIMA, and Holt–Winters methods. The objective is to assess their forecasting performance better to anticipate the economic impacts of stock price trends.	ARIMA, SutteArima, dan Holt-Winters	2019-2020
(Mayatopani, 2021)	This review assesses the predictive capability of neural networks for financial time series by comparing their forecasting accuracy with that of GARCH models. It also explores optimal neural network architectures across different data samples to enhance prediction performance.	ANN, GARCH	No mention.
(A. D. Hartanto et al., 2023)	This review aims to forecast stock price fluctuations using the LightGBM algorithm to achieve high accuracy and to compare its performance with that of other forecasting models.	LightGBM, GSCV, XGBoost, AdaBoost, CatBoost	1992-2022
(Zein et al., 2024)	This study aims to evaluate the N-BEATS model for stock price prediction accuracy, compare its performance with LSTM architectures, and support informed investment decisions in the Indonesian stock market.	N-BEATS, LSTM	2013-2023
(Kristiyanti et al., 2024)	This review examines the impact of inflation on investors' purchasing	LSTM	2011-2023



	power by focusing on stock price forecasting in Indonesia's transportation sector. The study applies data mining techniques within the CRISP-DM framework to enable systematic, structured analysis.		
(S. Hartanto et al., n.d.)	This review aims to enhance stock market prediction using deep learning models by investigating the integration of technical indicators into the Temporal Fusion Transformer (TFT). The study explores forecasting strategies to generate more accurate insights for informed investment decision-making.	LSTM, GRN	2019-2024
(Munthe et al., 2024)	This review aims to improve the accuracy of stock market trend prediction by applying advanced techniques and addressing class imbalance in financial datasets using SMOTE. It emphasizes integrating machine learning and deep learning models within a stacked framework to enhance predictive performance.	LSTM, XGBoost, Random Forest, Logistic Regression, SMOTE	No mention.
(Sadewa et al., 2025)	This study aims to develop a stock price prediction system for Netflix by evaluating the accuracy of the Simple Exponential Smoothing and Weighted Moving Average methods. The research is motivated by the increasing public interest in stock investment, highlighting the need for reliable forecasting models to support data-driven investment decision-making.	SES, WMA	2019-2024
(Gunaryati et al., 2025)	This review aims to improve the accuracy of Sharia-compliant stock price forecasting by proposing a hybrid ARIMA-LSTM model integrated with LOWESS regression. The study addresses the challenges of	ARIMA-LSTM	No mention.



	high volatility and non-linear patterns inherent in stock price data.		
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3.1 Discussion

Regarding RQ1, the reviewed studies demonstrate a strong dominance of time-series-based predictive models in stock price forecasting. Traditional statistical approaches such as ARIMA, GARCH, and their variants remain relevant, particularly for modeling volatility and risk, as evidenced by studies focusing on Value-at-Risk (VaR) and return volatility (Mayatopani, 2021; Tarno et al., 2021). These methods are often favored for their mathematical interpretability and effectiveness in relatively stable market conditions.

However, the literature increasingly highlights the superior performance of deep learning models, especially Long Short-Term Memory (LSTM) networks, in capturing non-linear patterns and long-term dependencies in financial time-series data. Several studies emphasize the effectiveness of LSTM in predicting stock prices across different sectors and markets, including Indonesian stocks and sector-specific indices (Budiharto, 2021; Hansun & Young, 2021; Kristiyanti et al., 2024). The recurrent structure of LSTM allows it to retain historical information more effectively than traditional models, making it particularly suitable for volatile market environments.

In addition to standalone deep learning models, hybrid and ensemble approaches are increasingly explored to enhance predictive accuracy. For instance, hybrid ARIMA–LSTM models are proposed to combine the strengths of linear statistical modeling and nonlinear deep learning, especially for Sharia-compliant stock forecasting (Gunaryati et al., 2025). Similarly, stacking frameworks that integrate machine learning and deep learning algorithms, supported by techniques such as SMOTE to address class imbalance, demonstrate improved predictive performance (Munthe et al., 2024). These findings indicate a clear trend toward methodological integration to overcome the limitations of single-model approaches.

Moreover, recent studies also explore advanced architectures beyond conventional LSTM models. Approaches such as N-BEATS (Zein et al., 2024), LightGBM and other gradient boosting methods (A. D. Hartanto et al., 2023), and Temporal Fusion Transformer (TFT) models (S. Hartanto & Gunawan, 2024) reflect ongoing efforts to improve forecasting accuracy and interpretability. Collectively, these results suggest that while LSTM remains the most frequently used method, research is gradually shifting toward more sophisticated and hybrid AI-based solutions.

With respect to RQ2, the findings indicate that historical stock price data is the primary information source used in most predictive models. Many studies rely on publicly available financial data platforms, with Yahoo Finance being the most frequently referenced source due to its accessibility and comprehensive historical records (Budiharto, 2021; Hansun & Young, 2021; Zein et al., 2024). Such data typically include daily opening, closing, high, low prices, and trading volume, which form the core inputs for time-series forecasting models.

Although historical price data dominate the reviewed literature, some studies acknowledge the potential benefits of incorporating additional information sources. For example, inflation indicators are used to analyze purchasing power and sector-specific stock behavior, particularly in transportation stocks (Kristiyanti et al., 2024). However, the integration of alternative data sources, such as news sentiment, social media signals, or macroeconomic indicators, remains limited in the reviewed studies, suggesting an opportunity for further exploration in future research.

This reliance on numerical price-based data reflects a broader trend in stock prediction research, in which quantitative indicators are preferred for their structured nature and ease of integration into machine learning pipelines. Nevertheless, the limited use of qualitative or sentiment-based data highlights a gap in current research, particularly in capturing market psychology and external shocks that influence stock price movements.

In addressing RQ3, the reviewed studies consistently employ error-based performance metrics to evaluate predictive models. Metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) are commonly used to assess forecasting accuracy across different modeling approaches (Budiharto, 2021; A. D. Hartanto et al., 2023; Zein et al., 2024). These metrics provide a straightforward means of comparing predicted values with actual stock prices and remain widely accepted in time-series forecasting research.

Deep learning-based studies, particularly those utilizing LSTM and hybrid architectures, frequently report improved performance in terms of lower prediction errors when compared to traditional statistical models (Gunaryati et al., 2025; Hansun & Young, 2021). This reinforces the conclusion that LSTM-based models are more effective in handling complex temporal dependencies inherent in stock price data.

Nevertheless, while these evaluation metrics effectively quantify predictive accuracy, they offer limited insight into model robustness, stability, and interpretability. Only a few studies extend evaluation beyond standard error metrics, such as those focusing on risk estimation using VaR or volatility modeling (Tarno et al., 2021). This indicates that future research could benefit from incorporating more comprehensive evaluation frameworks that consider both predictive accuracy and practical relevance for decision-making.

4. KESIMPULAN



This systematic literature review provides a comprehensive overview of recent research on stock price prediction using artificial intelligence methods. Based on the analysis of 13 selected studies, the findings reveal that time-series forecasting techniques remain the dominant approach in this field, with deep learning models, particularly LSTM and its hybrid variants, being the most widely adopted due to their ability to capture temporal dependencies and non-linear patterns in stock price data. In addition, machine learning models such as Random Forest, XGBoost, and LightGBM are increasingly utilized to enhance prediction accuracy, especially when integrated into ensemble or hybrid frameworks.

The review also indicates that historical stock price data obtained from platforms such as Yahoo Finance is the primary source of information used in predictive modeling. At the same time, only a limited number of studies incorporate alternative data such as sentiment analysis or macroeconomic indicators. Regarding model evaluation, most studies rely on standard error-based metrics, which remain effective for assessing predictive performance but offer limited insights into model robustness and interpretability.

Despite notable progress in AI-based stock price prediction, several challenges persist, including data quality issues, limited model interpretability, and difficulty capturing external shocks such as economic crises or geopolitical events. Therefore, future research is encouraged to explore more explainable AI models, incorporate diverse data sources, and develop adaptive learning frameworks capable of responding to rapidly changing market conditions. Overall, this study contributes to the existing body of knowledge by synthesizing recent advancements and identifying research directions that can support more accurate and reliable stock market forecasting models.

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